

# SPEC-2 - NeuroPathway API and Workflow Architecture

Next architecture section after the database and MVP schema work.

## Purpose

The database design defines what exists. SPEC-2 defines how the child app, adult dashboard, AI pattern reader, safeguarding engine, and database work together end-to-end.

## Current completed architecture

1. Database choice
2. Multi-tenant PostgreSQL model
3. Child-facing Pattern Recognition AI Reader concept
4. MVP database schema
5. Future spec roadmap

## Next item on the list

The next item is the API and workflow architecture, beginning with the most important endpoint: POST /v1/child/checkins.

```
Child submits check-in
-> API authenticates child and resolves tenant
-> PostgreSQL stores check-in
-> Safety pre-screen runs
-> AI extracts structured observations
-> Pattern engine updates behaviour patterns
-> Policy engine decides what the child can see
-> Adult review queue is created if needed
-> Safeguarding workflow triggers if high-risk
-> Audit event is written
```

## SPEC-2 contents

| Section                              | What it defines                                                         |
|--------------------------------------|-------------------------------------------------------------------------|
| API architecture                     | How clients, backend services, AI services, and databases connect       |
| Authentication and tenant resolution | How the request is tied to a tenant, user, child, and role              |
| Child-facing endpoints               | Check-ins, reflections, help requests, patterns                         |
| Adult/professional endpoints         | Review queue, child pattern view, decisions, support planning           |
| Internal AI endpoints                | Safety screening, extraction, pattern generation, reflection generation |
| Safeguarding endpoints               | Escalations, events, review outcomes                                    |
| Audit logging rules                  | Every sensitive read, write, AI action, export, and escalation          |
| Request/response examples            | Concrete JSON contracts for developers                                  |
| Sequence diagrams                    | End-to-end implementation flow                                          |
| Safety fallbacks                     | What happens when AI, queues, or external services fail                 |

## Primary endpoint

The first endpoint to design is:

POST /v1/child/checkins

## Responsibilities of POST /v1/child/checkins

- Validate authentication and child session
- Resolve tenant\_id, child\_id, and allowed role
- Validate structured check-in payload
- Store the raw child-authored check-in in PostgreSQL
- Write an audit event for the submission
- Run safety pre-screen before generating any reflection
- Trigger safeguarding workflow immediately for high-risk content
- Extract structured AI observations for low/medium-risk content
- Update behaviour patterns
- Generate child-safe reflection where policy allows
- Create adult review item where policy requires
- Return a safe child-facing response

## Draft child check-in request

POST /v1/child/checkins  
Authorization: Bearer <child\_session\_token>  
X-Tenant-ID: <tenant\_id>

```
{
  "checkin_type": "school_day",
  "mood_score": 2,
  "energy_score": 2,
  "sensory_score": 5,
  "stress_score": 4,
  "safety_score": 3,
  "location_context": "school",
  "activity_context": "lunch",
  "selected_feelings": ["angry", "worried"],
  "selected_body_signals": ["shaky", "tense"],
  "selected_triggers": ["noise", "crowd"],
  "selected_helpers": ["quiet_space", "trusted_adult"],
  "free_text": null
}
```

## Draft child-safe response

```
{
  "checkin_id": "uuid",
  "status": "received",
  "child_message": {
    "type": "gentle_reflection",
    "text": "Lunch felt hard today. A quiet space or trusted adult may help."
  },
  "pattern_update": {
    "new_pattern_detected": true,
    "child_safe_summary": "Busy lunch spaces may be difficult when they are noisy."
  },
  "adult_review_created": true,
  "safeguarding_escalation_created": false
}
```

## High-risk fallback response

```
{
  "checkin_id": "uuid",
  "status": "received",
  "child_message": {
    "type": "support_prompt",
    "text": "Thank you for telling us. Please speak to your trusted adult now."
  },
  "pattern_update": null,
  "adult_review_created": true,
  "safeguarding_escalation_created": true
}
```

## Initial API groups

### Child-facing app

```
POST /v1/child/checkins
GET  /v1/child/reflections
GET  /v1/child/patterns
POST /v1/child/help-request
```

### Adult/professional dashboard

```
GET  /v1/children/{child_id}
GET  /v1/children/{child_id}/patterns
GET  /v1/adult/review-queue
POST /v1/adult/reviews/{review_id}/decision
POST /v1/safeguarding/events
```

### Internal AI orchestration

```
POST /v1/internal/ai/safety-screen
POST /v1/internal/ai/extract-observations
POST /v1/internal/ai/generate-patterns
POST /v1/internal/ai/generate-child-reflection
```

### Audit

```
POST /v1/internal/audit/events
GET  /v1/audit/children/{child_id}
```

## Safety principles for SPEC-2

- No child-facing AI output before safety screening
- No diagnostic or risk labels shown to the child
- No high-risk interpretation shown directly to the child
- Medium-risk content creates adult review
- High-risk content creates safeguarding workflow
- Every AI output includes source evidence, model version, prompt version, confidence, and audience
- Every sensitive action creates an audit event
- Tenant isolation is enforced in application logic and PostgreSQL Row-Level Security

## After SPEC-2

```
SPEC-3 - Safeguarding Workflow
SPEC-4 - AI Governance and Safety
SPEC-5 - Frontend UX for Children
SPEC-6 - Infrastructure and Deployment
SPEC-7 - Implementation Plan and Milestones
```

## Decision

Proceed next with SPEC-2: API and Workflow Architecture, starting with POST /v1/child/checkins as the anchor flow. Once this is defined, the frontend, backend, AI orchestration layer, safeguarding workflow, and audit model can be implemented coherently by contractors.